

TLA+ Study Group — Week 02

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Lecture objective

Try to model the problem of filling the jugs with water within TLA+.

What stood out

- Allowing only behaviours which allow where exactly how much water is in the jugs at any given time.
- Going through jug example makes the reader learn what the variables and steps are.
 - There are no intermediate states, filling a jug is one step. Classic where impartial states are impossible.
 - TLA does not have type declarations. *TypeOK* example shows that every type can be typechecked with TLC.

$$\begin{aligned} TypeOK &\triangleq \wedge \ small \in 0..3 \\ &\wedge \ big \in 0..5 \end{aligned}$$

- *TypeOK* is not considered part of the Spec.
- It feels that **writing** the spec in TLA+ is simpler than it would be to solve this in programming language. Just add invariants and TLC solves it.
- *At the moment*, however, finding bugs in your spec seems harder than it would be in programming language. It seems easy to write a bad spec, hoping to learn how to avoid this.

Questions

- How to model states which require intermediate states due infrastructure concerns: e.g. transactionality and telemetry. Do we make our states represent this as well?